

BGFM-NATIVE: ENERGY-COUPLED BOLTZMANN FLOW MATCHING WITH UNIVERSAL NEURAL POTENTIALS FOR 3D MOLECULAR GENERATION

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Paper under double-blind review

ABSTRACT

Fast 3D molecular flow and diffusion models generate diverse de novo molecules, but their probability geometry is learned from empirical molecular datasets rather than from a physical energy landscape. Energy-based molecular generators recover an explicit scalar energy, but sample with expensive Langevin or parallel-tempering chains and can trade diversity for physical quality. We propose **BGFM**, an energy-coupled flow-matching generator for de novo 3D molecular generation. Its coordinate velocity is decomposed into an amortized transport field and a conservative Boltzmann residual drift distilled from OMol25-derived energies and forces (Levine et al., 2025): $v_{\text{BGFM}} = v_{\theta} + \alpha(t, T_{\text{drift}}) P_{\text{SE}(3)}[-\beta_{\text{drift}} \nabla_r \Delta E_{\psi}]$. Training couples this vector field to the external energy landscape through late-time force-score consistency, within-parent density-energy variance, and a finite-cloud Local Boltzmann bridge. BGFM also learns a composition-conditioned residual strain head from the same teacher and uses it inside a short coordinate-only corrector that is unrolled during training. The resulting model is designed to combine the speed and diversity of flow matching with the low-strain sampling behavior of energy-based models. Because OMol25 labels enter training, we separate mechanistic OMol25 alignment from external validation: the headline experiments use standard QM9 and GEOM-Drugs (Axelrod and Gómez-Bombarelli, 2022) generation metrics, independent GFN2-xTB (Bannwarth et al., 2019), MMFF (Halgren, 1996), and DFT relaxation oracles, quality-diversity-compute Pareto curves with the Vendi score (Friedman and Dieng, 2023), and shuffled-label negative controls.

1 INTRODUCTION

Three-dimensional molecular generators are increasingly used as the front end of simulation and discovery pipelines. A useful generated structure must not only be chemically valid; it should already be close to a physically plausible region of an energy landscape, otherwise substantial relaxation or molecular dynamics is required before the sample can be used. Current models expose a tradeoff. Flow-matching and diffusion models amortize sampling and produce diverse molecules, but standard training matches empirical data distributions and does not, by construction, enforce Boltzmann relative probabilities under a quantum-level energy model. Energy-based molecular generators learn an explicit scalar potential and can rank, filter, and steer samples, but sampling from a learned energy generally requires many inference-time steps.

BGFM addresses this gap by treating a universal neural potential as an external physical teacher for the generative vector field itself. We instantiate the amortized transport component with a FlowMol3-style de novo proposal (Dunn and Koes, 2025) over atom types, charges, and coordinates, but the coordinate velocity used for training and sampling is energy-coupled: it adds a learned conservative residual drift calibrated from OMol25. The first calibration term aligns the flow-implied coordinate score with neural-potential forces near the data endpoint. The second estimates a coordinate log-density with a FFJORD integral and enforces that, within a cloud of perturbations of the

same parent molecule,

$$\log p_{\theta}^{\text{pos}}(r | c) + \beta_{\text{phys}} E_{\text{NP}}(r, c) \quad (1)$$

varies as little as possible, where $\beta_{\text{phys}} = 1/(k_B T_0)$ is the physical inverse temperature (see ‘‘Temperature conventions’’ in §2’s preamble). This is a local conditional Boltzmann condition: lower-energy perturbations of the same molecular identity should receive higher probability. BGFM further trains a small scalar energy head to approximate the external neural-potential energy and force, then uses this head for a short learned-energy Langevin corrector that refines coordinates while holding the sampled molecular identity fixed.

The central comparison is with EBMol-style energy-based generation (Griesbacher et al., 2026). EBMol learns a persistent scalar energy and samples with Mirror-Langevin dynamics and parallel tempering. BGFM takes the complementary route: keep the fast flow proposal, calibrate it with external physical labels, and use an energy-coupled flow field plus a short learned-energy corrector rather than a long MCMC chain. Therefore, our primary evaluation is not just validity or an OMol25 self-correlation. We evaluate whether BGFM improves the physical-quality–diversity–compute frontier under independent relaxation oracles.

Contributions.

- Energy-coupled equivariant molecular vector field.** We design an SE(3)-equivariant generative velocity field that decomposes into an amortized transport component and a conservative late-time Boltzmann drift distilled from a universal neural potential, $v_{\text{BGFM}} = v_{\theta} + \alpha(t, T_{\text{drift}}) P_{\text{SE}(3)}[-\beta_{\text{drift}} \nabla_r \Delta E_{\psi}]$. The energy gradient enters the generator’s dynamics, not a post-processing step.
- Local Boltzmann bridge with density-energy calibration.** We train the position log-density to match a finite- cloud Boltzmann target induced by OMol25 perturbation energies via an explicit per-parent KL bridge, $\text{KL}(w_m \| q_{\theta}^{(m)})$ with $w_m^{(k)} \propto e^{-\beta_{\text{bridge}}(s) E_m^{(k)}}$ and $q_{\theta}^{(m,k)} \propto e^{\log p_{\theta}(r_m^{(k)} | c_m)}$, in parallel with the within-parent density-energy variance form. Here $\beta_{\text{bridge}}(s) = 1/(k_B T_{\text{bridge}}(s))$ is the *tempered* training inverse temperature, annealed to the physical anchor β_{phys} by end of training. Both objectives are invariant to the unknown per-composition partition function and reuse the same FFJORD log-density evaluation.
- Corrector-in-the-loop amortized physical sampler.** A composition-conditioned residual strain scorer $\hat{E}_{\psi}(r, c) = b_{\phi}(c) + \Delta E_{\psi}(r, c)$ is calibrated to OMol25 force/energy. We unroll a random short Langevin corrector on this learned energy at training time with a stability penalty, so the sampler obtains EBM-like physical refinement without long Langevin or parallel-tempering chains and without a separate post-processing stage.
- Quality–diversity–compute evaluation.** Headline metrics use independent oracles (GFN2-xTB and MMFF relaxation, DFT subset) rather than OMol25 self-scores; we report a Pareto frontier across model variants alongside ablations and negative controls that isolate the contribution of each module.

2 RELATED WORK

3D molecular flow and diffusion. EDM (Hoogeboom et al., 2022), GeoLDM (Xu et al., 2023), MiDi (Vignac et al., 2023), GeoBFN (Song et al., 2024), SemlaFlow (Irwin et al., 2024), FlowMol (Dunn and Koes, 2024), and FlowMol3 (Dunn and Koes, 2025) learn transport or denoising maps for molecular atom types and coordinates. GeoDiff (Xu et al., 2022) targets conditional conformer generation. The underlying generative families — denoising diffusion (Ho et al., 2020), score SDE (Song et al., 2021), neural ODEs and FFJORD-style flow density (Chen et al., 2018; Grathwohl et al., 2019), and flow matching (Lipman et al., 2023) — supply the velocity / score primitives we reuse. FlowMol3 is the closest same-family baseline: an open-source multi-modal flow-matching model for de novo all-atom small-molecule generation that improves FlowMol via self-conditioning, fake atoms, and train-time geometry distortion without changing the underlying GNN or flow formulation. BGFM does not replace the discrete molecular identity generator and does not enlarge the GNN backbone; instead, it changes the *coordinate dynamics* into an energy-coupled flow field. Concretely, the continuous coordinate vector field is modified into an energy-coupled vector field

whose conservative component is distilled from a universal neural potential, while the amortized molecular identity proposal over atom types and charges is inherited from FlowMol3. The change is to the generative dynamics themselves, not to a post-hoc calibration layer applied to the backbone’s outputs.

Energy-based molecular generation. EBMol (Griesbacher et al., 2026) learns a time-unconditional atom-additive scalar potential and samples it with Mirror-Langevin dynamics plus parallel tempering. It is therefore the direct competitor for physically motivated 3D generation. BGFM differs in two ways. First, BGFM uses an external universal neural potential as the physical teacher, whereas EBMol learns an energy from geometry data without direct energy supervision. Second, BGFM retains an amortized flow proposal and uses a short corrector, whereas EBMol relies on inference-time energy sampling.

Flexible conformer ensemble generation. FlexiFlow (Tedoldi et al., 2025) extends flow matching to jointly sample molecular graphs and multiple conformations and reports strong QM9/GEOM-Drugs performance. FlexiFlow is a required standard-generation baseline, especially for conformer-ensemble metrics. Its goal is complementary to BGFM: FlexiFlow focuses on molecular/conformer ensemble coverage, while BGFM focuses on energy-coupled coordinate dynamics that produce low-strain de novo geometries without long inference-time energy chains.

Boltzmann generators and energy-based flow matching. Boltzmann Generators (Noé et al., 2019), equivariant flow matching for Boltzmann sampling (Klein and Noé, 2023), and iEFM (Woo and Ahn, 2024) study sampling from known or learned unnormalized densities, often for fixed systems or continuous toy/physical potentials. iEFM is especially relevant theoretically because it trains CNFs from energy function evaluations, but its evaluation setting is fixed continuous Boltzmann sampling rather than mixed discrete-continuous de novo molecule generation.

Universal neural potentials. OMol25 (Levine et al., 2025) provides large-scale DFT data and neural-potential models over a broad chemistry space. BGFM uses such a neural potential not as a post-hoc relaxer but as a training-time teacher for forces, energies, and energy-head calibration.

3 METHOD

Temperature conventions. Throughout this paper we use a strict two-temperature convention. The *physical* temperature is $T_0 = 298.15$ K and the corresponding physical Boltzmann constant times temperature is $k_B T_0 = 0.02569$ eV = 0.593 kcal/mol, so $\beta_{\text{phys}} = 1/(k_B T_0) \approx 38.93$ eV⁻¹. This is the unique definition of “room temperature” used to define the target Boltzmann law and the reported physical-quality numbers (the xTB/MMFF/DFT ΔE tables and BGFM-Native-det). During training, the Boltzmann-bridge loss uses an *effective* $k_B T_{\text{bridge}}(s)$ that may be tempered or annealed for numerical stability of the variance / KL objective; whenever $k_B T_{\text{bridge}} > k_B T_0$ we call this configuration the *tempered Boltzmann bridge* and never “room temperature”. We recommend a log-linear anneal $0.25 \rightarrow 0.10 \rightarrow 0.05 \rightarrow 0.02569$ eV. The corrector is reported in two variants (§3.5): a *deterministic* descent at T_0 and a *physical-300 K noisy Langevin* diagnostic that uses β_{phys} for its noise scale. All “ kT ”, “ T_0 ”, “298”, and “tempered” usage in the paper is labelled in one of these three categories. Any kT used in training that differs from $k_B T_0$ is the *effective / tempered training temperature*, not physical.

Let a molecular configuration be $x = (c, r)$, where c denotes molecular identity variables (atom count, atom types, and charges) and $r \in \mathbb{R}^{3N}$ denotes coordinates. The external neural potential returns an energy $E_{\text{NP}}(r, c)$ and forces $F_{\text{NP}}(r, c) = -\nabla_r E_{\text{NP}}(r, c)$. BGFM does not define a global canonical ensemble over arbitrary molecular identities. Instead, for a fixed identity c , it targets the conditional coordinate Boltzmann law

$$p_B(r \mid c, T_0) = \frac{1}{Z_c(T_0)} \exp[-\beta_{\text{phys}} E_{\text{NP}}(r, c)], \quad \beta_{\text{phys}} = 1/(k_B T_0) \text{ [physical]}. \quad (2)$$

This target is well-defined because c is fixed; cross-composition probabilities are left to the empirical proposal distribution and are not interpreted as thermodynamic weights.

3.1 ENERGY-COUPLED FLOW PROPOSAL

BGFM uses a FlowMol3-style proposal for molecular identity variables, but the coordinate velocity is an explicitly energy-coupled vector field. Let $v_\theta(x_t, t)$ be the amortized transport field from the flow proposal and let $\Delta E_\psi(r, c)$ be a composition-conditioned residual strain energy head. The coordinate velocity used by BGFM is

$$v_{\text{BGFM}}(x_t, t) = v_\theta(x_t, t) + \alpha(t, T_{\text{drift}}) P_{\text{SE}(3)}[-\beta_{\text{drift}} \nabla_r \Delta E_\psi(r_t, c)], \quad \beta_{\text{drift}} = 1/(k_B T_{\text{drift}}), \quad (3)$$

where $P_{\text{SE}(3)}$ removes per-graph center-of-mass drift and $\alpha(t, T_{\text{drift}})$ is a late-time onset schedule. The explicit β_{drift} makes the role of temperature in the drift unambiguous and matches Eq. equation 13. Atom types and charges still use the categorical flow/CTMC component of the proposal. The base flow-matching loss is applied to the total velocity,

$$\mathcal{L}_{\text{FM}} = \mathbb{E} [\|v_{\text{BGFM}}(x_t, t) - u_t\|^2 + \mathcal{L}_{\text{disc}}]. \quad (4)$$

This turns the external energy from a post-hoc correction signal into part of the generative dynamics while retaining amortized flow sampling. In ablations we also report the plain-flow proposal v_θ and a post-hoc-only variant.

3.2 LATE-TIME FORCE-SCORE CONSISTENCY

For the conditional Boltzmann law in Eq. equation 2,

$$\nabla_r \log p_B(r | c, T_0) = \beta_{\text{phys}} F_{\text{NP}}(r, c) \quad [\text{physical}]. \quad (5)$$

Under the linear Gaussian interpolant, the coordinate score at time t can be estimated from the flow velocity by

$$s_\theta(x_t, t) = \frac{t v_\theta(x_t, t) - x_t}{(1 - t)\sigma_0^2}. \quad (6)$$

BGFM evaluates this score at late but finite times $t \in \mathcal{T}$ and aligns it with the endpoint neural-potential force:

$$\mathcal{L}_{\text{force}} = \mathbb{E}_{t \in \mathcal{T}} [1 - \cos(s_\theta(x_t, t), \beta_{\text{phys}} F_{\text{NP}}(r_1, c))] \quad [\text{physical}; \cos \text{ is } \beta\text{-invariant}]. \quad (7)$$

This is a late-time regularizer, not an exact identity at arbitrary intermediate x_t . It avoids online neural-potential calls during every training step by using precomputed endpoint forces.

3.3 DENSITY-ENERGY VARIANCE

The most direct probability calibration is that $\log p_B(r | c, T) + \beta_{\text{bridge}}(s) E_{\text{NP}}(r, c)$ equals a constant depending only on c and T . BGFM estimates the coordinate log-density $\log p_\theta^{\text{pos}}(r | c)$ with a FFJORD-style divergence integral through the coordinate velocity field. For each parent molecule m , we precompute a cloud of perturbations $\{r_m^{(k)}\}_{k=1}^K$ and neural-potential energies. The loss is

$$\mathcal{L}_{\text{dens}} = \mathbb{E}_m \text{Var}_k [\log p_\theta^{\text{pos}}(r_m^{(k)} | c_m) + \beta_{\text{bridge}}(s) E_{\text{NP}}(r_m^{(k)}, c_m)], \quad (8)$$

where $\beta_{\text{bridge}}(s)$ is the (possibly tempered) training inverse temperature that anneals to β_{phys} by end of training (see ‘‘Temperature conventions’’). Because the variance is taken within the same parent molecule, the unknown offset $-\log Z_{c_m}(T)$ cancels. We optionally add a composition-conditioned scalar offset $b_\phi(c)$ as an anchor, but do not interpret it as a supervised thermodynamic partition-function estimator.

3.4 CALIBRATED COMPOSITION-CONDITIONED STRAIN SCORER

To obtain an explicit inference-time energy without running OMol25 at sampling time, BGFM trains a scalar head. Absolute OMol25 energies are dominated by composition and atom count, so a direct regression head would mostly learn molecular size rather than coordinate-dependent strain. We therefore decompose the head into a *composition baseline* $b_\phi(c)$ and a *coordinate-dependent strain term* $\Delta E_\psi(r, c)$:

$$\hat{E}_\psi(r, c) = b_\phi(c) + \Delta E_\psi(r, c), \quad (9)$$

and train against OMol25 with the residual loss

$$\mathcal{L}_{\text{head}} = |\Delta E_{\psi}(r, c) - (E_{\text{NP}}(r, c) - b_{\phi}(c))| + \lambda_F \|\nabla_r \Delta E_{\psi}(r, c) - F_{\text{NP}}(r, c)\|_2^2, \quad (10)$$

where b_{ϕ} is a small invariant network on the molecular composition vector (atom-type histogram + total charge) and ΔE_{ψ} is the per-graph readout of the head from the flow-proposal scalar features. We use the head as a composition-conditioned strain scorer for ranking and refinement; it is not a free energy and is not a thermodynamic absolute formation energy.

3.5 SHORT LEARNED-ENERGY CORRECTOR

After the flow proposal samples (c, r^0) , BGFM optionally refines only coordinates using the calibrated head. We report two variants:

Deterministic descent corrector (BGFM-Native-det).

$$r^{j+1} = \Pi[r^j + \eta_j(-\nabla_r \Delta E_{\psi}(r^j, c))], \quad j = 0, \dots, J-1. \quad (11)$$

This is the noise-free ($\epsilon_j = 0$) descent that produces our *main physical-quality numbers* (xTB/MMFF ΔE). It is not intended as a Boltzmann sampler; it is a short, learned, deterministic relaxation, comparable to a few steps of MMFF or xTB local optimization but using only the in-model strain head.

Room-temperature noisy Langevin corrector (BGFM-Native-300K).

$$r^{j+1} = \Pi[r^j + \eta_j(-\nabla_r \Delta E_{\psi}(r^j, c)) + \sqrt{2\eta_j/\beta_{\text{phys}}} \epsilon_j], \quad \epsilon_j \sim \mathcal{N}(0, I), \quad (12)$$

with $\beta_{\text{phys}} = 1/(k_B T_0)$ at $T_0 = 298.15$ K [physical]. This is the mechanistic Boltzmann-ensemble diagnostic and is what we use when we wish to test conditional-Boltzmann calibration empirically; it is *not* the default reported number. Here $J \in \{0, 20, 50, 100\}$ in our compute sweep, Π removes center-of-mass drift, and the discrete identity c — atom counts, types, and charges — is held fixed throughout refinement. Per sampled molecule we report flow NFEs, energy-head NFEs, external oracle calls (zero by default), wall-clock time, and outcome. Samples whose post-corrector geometry produces invalid OpenBabel/RDKit bond inference, steric clash above a threshold, or downstream relaxation failure are counted as *failures* and remain in the population averages of the all-samples protocol; we never silently drop them. Unless a Metropolis correction is explicitly enabled and reported, neither corrector variant is an exact MCMC sampler.

3.6 ENERGY-CALIBRATED VECTOR FIELD

Modules 1–4 above use the energy head only at sampling time. To turn BGFM into a genuine *energy-calibrated* flow rather than a flow with a post-hoc relax step, we additionally inject the head’s coordinate gradient into the velocity field itself, with an explicit inverse-temperature factor $\beta_{\text{drift}} = 1/(k_B T_{\text{drift}})$ [physical at default]:

$$v_{\text{BGFM}}(x_t, t) = v_{\theta}(x_t, t) + \alpha(t, T_{\text{drift}}) P_{\text{SE}(3)}[-\beta_{\text{drift}} \nabla_r \Delta E_{\psi}(r_t, c)], \quad (13)$$

where $\alpha(t, T_{\text{drift}})$ is a smooth late-time onset schedule (zero on $[0, t_{\text{on}}]$, growing on $[t_{\text{on}}, 1]$ with a quadratic power) and $P_{\text{SE}(3)}$ removes the per-graph centroid drift so the correction respects translation equivariance; $\text{SO}(3)$ equivariance is inherited from the strain term being a rotation-invariant scalar of pairwise distances. The drift inverse temperature β_{drift} appears *explicitly* in the velocity so the role of temperature in the generative dynamics is unambiguous; in the main configuration $k_B T_{\text{drift}} = k_B T_0 = 0.02569$ eV [physical], and the temperature ablation in §4.5 sweeps it. We add a stability regularizer

$$\mathcal{L}_{\text{drift}} = \mathbb{E} \|\alpha(t, T_{\text{drift}}) P_{\text{SE}(3)}[-\beta_{\text{drift}} \nabla_r \Delta E_{\psi}(r_t, c)]\|^2, \quad (14)$$

so the drift contribution remains the same order of magnitude as the base velocity. The norm penalty alone, however, does not certify the *direction* of the drift: a backbone that learned to cancel the drift contribution would still satisfy Eq. equation 14 at the optimum. To make the drift a genuinely

physical *descent* direction on the learned strain, we add a hinge *monotonicity* loss. Writing $d_t = P_{SE(3)}[-\beta_{\text{drift}} \nabla_r \Delta E_\psi(r_t, c)]$ for the conservative drift component, the loss is

$$\mathcal{L}_{\text{mono}} = \mathbb{E} \max(0, \Delta E_\psi(r_t + \epsilon d_t, c) - \Delta E_\psi(r_t, c) + m), \quad (15)$$

with step size ϵ matched to the late-time velocity scale and margin $m \geq 0$. The role of m is to interpolate between two regimes: $m = 0$ asks only that a step along d_t not *raise* the learned strain (so a flat optimum where $d_t \approx 0$ satisfies the hinge); $m > 0$ asks that the strain strictly decrease by at least m along d_t , which forbids the flat optimum and forces d_t to be a non-trivial descent direction whenever ΔE_ψ has a non-zero gradient at r_t . We default to $m = 0$ in the main configuration because the early-training strain head is noisy and a strict-descent margin amplifies that noise into instability; we sweep $m \in \{0, 0.05, 0.1\}$ eV in the ablation table. With either margin, this hinge certifies that the conservative component, when actually applied, does not raise the learned strain on average and thus distinguishes a physical Boltzmann drift from a generic norm-controlled perturbation. At inference, the sampler can opt in to either the plain flow v_θ or the calibrated field v_{BGM} ; we report both in the experimental ablations.

3.7 CORRECTOR-AWARE TRAINING

The post-hoc Langevin corrector (§3.5) can be criticized as a generic post-processing step accessible to any amortized baseline. To close that gap, we train the energy head and the flow jointly with a random short unroll of the corrector: sample $J \sim \text{Uniform}\{0, 1, \dots, J_{\text{max}}\}$ with $J_{\text{max}} = 3$, run J Langevin steps from the data endpoint under $\hat{E}_\psi$, and add a stability penalty

$$\mathcal{L}_{\text{stab}} = \mathbb{E}_{\text{atom}} [\max(0, \|r' - r_1\|^2 - \delta_{\text{max}}^2)] + \mathbb{E}_{\text{graph}} [\max(0, \hat{E}_\psi(r', c) - \hat{E}_\psi(r_1, c))], \quad (16)$$

where r' is the unrolled geometry and δ_{max} caps the per-atom displacement. This makes the corrector an in-the-loop training signal rather than a sampling-only procedure, at a cost of roughly J_{max} additional energy-head forward and backward passes per batch.

3.8 LOCAL BOLTZMANN BRIDGE

The density-energy variance objective in Eq. equation 8 controls only the second moment of the residual across each parent’s perturbation cloud. To turn that into a direct distribution-matching objective with a well-defined target, we add a Local Boltzmann bridge loss. For each parent molecule m with K perturbations, define the neural-potential-induced cloud-restricted Boltzmann target

$$w_m^{(k)} = \frac{\exp[-\beta_{\text{bridge}}(s) E_{\text{NP}}(r_m^{(k)}, c_m)]}{\sum_{j=1}^K \exp[-\beta_{\text{bridge}}(s) E_{\text{NP}}(r_m^{(j)}, c_m)]}, \quad \beta_{\text{bridge}}(s) = 1/(k_B T_{\text{bridge}}(s)) \text{ [tempered, anneals to } \beta_{\text{phys}}], \quad (17)$$

and the model’s cloud-restricted distribution

$$q_\theta^{(m,k)} = \frac{\exp[\log p_\theta^{\text{pos}}(r_m^{(k)} | c_m)]}{\sum_{j=1}^K \exp[\log p_\theta^{\text{pos}}(r_m^{(j)} | c_m)]}. \quad (18)$$

The unknown partition function $-\log Z_{c_m}(T)$ enters both numerator and denominator equally inside each softmax and cancels exactly; the bridge loss is therefore well-defined despite $\log p_\theta^{\text{pos}}$ being only known up to a per-parent constant in practice. We optimize the per-parent forward KL,

$$\mathcal{L}_{\text{bridge}} = \mathbb{E}_m \text{KL}(w_m \| q_\theta^{(m)}) = \mathbb{E}_m \sum_{k=1}^K w_m^{(k)} (\log w_m^{(k)} - \log q_\theta^{(m,k)}). \quad (19)$$

In the small-fluctuation limit where the cloud residual $\log p_\theta^{\text{pos}} + \beta_{\text{bridge}}(s) E_{\text{NP}}$ is approximately constant across k , this reduces to half the within-parent variance in Eq. equation 8 plus higher-order terms, so the two objectives are theoretically consistent. In practice the bridge form is more numerically stable when the cloud is large or when the residual is not yet calibrated, and it is the form reviewers more readily recognize as an explicit distribution-matching algorithm. The bridge reuses the single FFJORD log-density evaluation already consumed by $\mathcal{L}_{\text{dens}}$ and thus carries no additional integration cost.

Symmetric variant. By default, $\mathcal{L}_{\text{bridge}}$ uses the forward $\text{KL}(w_m \| q_{\theta}^{(m)})$, which is mode-covering: the model is encouraged to place mass on every low-energy perturbation that OMol25 marks as probable. An alternative, made available as a config knob, is the symmetric Jensen-Shannon-style form

$$\mathcal{L}_{\text{bridge}}^{\text{sym}} = \frac{1}{2} \mathbb{E}_m [\text{KL}(w_m \| q_{\theta}^{(m)}) + \text{KL}(q_{\theta}^{(m)} \| w_m)], \quad (20)$$

which adds a mode-seeking term that penalizes the model for placing mass on cloud entries that OMol25 considers high-energy. We default to forward KL because the natural failure mode of an undertrained model is to assign uniform mass over the cloud (which forward KL strongly penalizes for low-energy targets); the symmetric variant is reported in the bridge ablation column of §4.6.

3.9 TRAINING OBJECTIVE

The full training loss is

$$\mathcal{L} = \mathcal{L}_{\text{FM}} + \lambda_1 \mathcal{L}_{\text{force}} + \lambda_2 \mathcal{L}_{\text{dens}} + \lambda_3 \mathcal{L}_{\text{anchor}} + \lambda_4 \mathcal{L}_{\text{head}} + \lambda_5 \mathcal{L}_{\text{drift}} + \lambda_6 \mathcal{L}_{\text{stab}} + \lambda_7 \mathcal{L}_{\text{bridge}} + \lambda_8 \mathcal{L}_{\text{mono}}. \quad (21)$$

All auxiliary weights use warmup/ramp schedules. OMol25 is not called in the main training loop: endpoint forces are stored in the dataset and perturbation energies are precomputed offline.

3.10 JOINT DISCRETE-CONTINUOUS DENSITY

BGFM exposes an optional *joint density* mode that augments the coordinate FJORD with the model’s discrete log-probability over atom counts N , atom types a , and per-atom formal charges c :

$$\log p_{\theta}(x | T) = \underbrace{\log p_{\theta}^{\text{pos}}(r | c, T)}_{\text{FJORD on coords}} + \underbrace{\log p_{\theta}^N(N) + \sum_{i=1}^N [\log p_{\theta}^a(a_i) + \log p_{\theta}^c(c_i)]}_{= \log p_{\theta}(c) \text{ (discrete CTMC)}}. \quad (22)$$

When enabled, $\log p_{\theta}(x | T)$ replaces $\log p_{\theta}^{\text{pos}}$ inside both the density-energy variance loss (Eq. equation 8) and the Local Boltzmann bridge target (Eq. equation 18). Because all K perturbations of a parent molecule m share atom counts, types, and charges, the discrete term contributes the same per-parent constant to every cloud entry and therefore drops out of both the within-parent variance and the per-parent softmax. The joint-density mode is consequently *invariant* for the parent-restricted objectives that we use as calibration signals, by construction. Its value is diagnostic: tracking the discrete component lets us audit whether the discrete generator is also implicitly Boltzmann-calibrated across compositions (a stricter condition than this paper claims) and is enabled in the λ_{joint} ablation column of §4.6. The headline experimental numbers in this paper use the conditional- coordinate variant ($\log p_{\theta}^{\text{pos}}$ only).

Algorithmic form. Training. For each minibatch, sample a flow time t and construct x_t on the conditional path. Evaluate v_{BGFM} in Eq. equation 3, compute the base flow-matching loss, the late-time force-score loss, FJORD log-densities on perturbation clouds, the variance and bridge losses, residual strain-head calibration, and a random 0 - J_{max} corrector unroll. Update all parameters with Eq. equation 21. **Sampling.** Integrate the learned energy-coupled vector field from the prior. Optionally run the same learned coordinate-only corrector for $J \in \{0, 20, 50, 100\}$ steps; no external OMol25 call is used in default sampling.

Proposition 1 (Finite-cloud Boltzmann ratios). *For a fixed molecular identity c , if $\log p_{\theta}^{\text{pos}}(r_m^{(k)} | c) + \beta_{\text{phys}} E_{\text{NP}}(r_m^{(k)}, c)$ is constant over the perturbation cloud (in the limit $T_{\text{bridge}} \rightarrow T_0$), then for any perturbations i, j ,*

$$\log \frac{p_{\theta}^{\text{pos}}(r_m^{(i)} | c)}{p_{\theta}^{\text{pos}}(r_m^{(j)} | c)} = -\beta_{\text{phys}} (E_{\text{NP}}(r_m^{(i)}, c) - E_{\text{NP}}(r_m^{(j)}, c)). \quad (23)$$

Thus, in the limit $T_{\text{bridge}} \rightarrow T_0$, the variance and bridge objectives enforce local conditional physical Boltzmann relative probabilities without requiring $Z_c(T_0)$.

Proposition 2 (Endpoint force-score condition). *Under the conditional coordinate Boltzmann target in Eq. equation 2, the coordinate score satisfies $\nabla_r \log p_B(r \mid c, T_0) = \beta_{\text{phys}} F_{\text{NP}}(r, c)$. Aligning the flow-implied late-time score of the total energy-coupled field with $\beta_{\text{phys}} F_{\text{NP}}$ is therefore a local necessary condition for physical Boltzmann calibration near the data endpoint.*

4 EXPERIMENTS

Experimental questions. We design experiments to answer six questions. First, does BGFM preserve standard de novo 3D generation quality? Second, does it improve physical quality under independent evaluators rather than only under the OMol25 labels used for training? Third, does it achieve a better quality-diversity-compute frontier than EBMol-style energy sampling? Fourth, does the proposed objective actually produce local conditional Boltzmann alignment? Fifth, do the gains require correct physical labels rather than arbitrary regularization? Sixth, does the calibrated energy head rank and filter molecular strain better than existing energy scorers?

Datasets and baselines. We evaluate on QM9, GEOM-Drugs, and held-out OMol25 chemistry slices. We report each baseline along with its post-processing variant, so the calibrated-corrector contribution of BGFM is not confounded with generic post-hoc refinement. Specifically, our standard-generation baselines include:

- the official FlowMol3 (Dunn and Koes, 2025) checkpoint with reported numbers and a re-run on our hardware;
- a *same-backbone FM baseline*: an identical FlowMol3-style architecture trained on the same OMol25 split, without any BGFM regularization or corrector;
- FlowMol3/FM *plus the same corrector budget* as BGFM, where the corrector uses an external xTB or OMol25 oracle instead of $\hat{E}_\psi$;
- FlowMol3/FM *plus an xTB / OMol25 re-ranking head* that filters and re-orders the same sample pool;
- EDM, GeoLDM, GeoBFN, SemlaFlow, FlexiFlow, and EBMol with reported numbers, run under matched NFE budgets where possible.

For BGFM we report the full method and its module ablations: flow-only, force-only, density-only, head-only, BGFM w/o corrector, and full BGFM. These splits let us answer whether BGFM’s gains come from the calibrated corrector or from post-processing accessible to any amortized baseline.

4.1 STANDARD GENERATION QUALITY

We generate 10k samples per model under matched or reported NFE budgets and report atom stability, molecule stability, validity, uniqueness, novelty, Vendi diversity, NFE, and wall-clock time.

Table 1: Standard de novo 3D generation quality. Replace all placeholders before submission.

Model	NFE	Atom stab.	Mol. stab.	Valid	Unique	Novel	Vendi
FlowMol3 (official)	reported	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
Same-backbone FM	[RUN: NFE]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
FlexiFlow	reported	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
EBMol	matched	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
BGFM w/o corrector	matched	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
BGFM full	matched	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]

4.2 INDEPENDENT PHYSICAL QUALITY

Because BGFM uses OMol25 labels during training, OMol25 correlation is not the headline metric. Our main physical-quality table uses independent GFN2-xTB relaxation, MMFF relaxation on organic subsets, and a DFT validation subset. We report both valid-connected-only and all-samples protocols.

Table 2: Independent physical quality under evaluators not used as BGFM’s density-energy target.

Model	Valid conn.	xTB med. ΔE	xTB mean ΔE	p90 ΔE	Relax steps	Failure
EBMol-low	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
EBMol-high	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
FlowMol3 (official)	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
Same-backbone FM	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
FM + corrector (same J)	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
FM + xTB rerank	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
BGFM w/o head	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
BGFM w/o corrector	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
BGFM full	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]

4.3 QUALITY-DIVERSITY-COMPUTE PARETO

We sweep flow NFEs and corrector steps $J \in \{0, 20, 50, 100\}$ and compare to EBMol’s compute-scaled samplers. The key metric is the number of valid, unique, low-strain samples per GPU-hour:

$$\text{VULS}_\tau = \frac{\#\{\text{valid, unique samples with } \Delta E_{\text{xTB}} < \tau\}}{\text{GPU-hour}}. \quad (24)$$

This evaluation targets the main BGFM hypothesis: a fast amortized flow proposal plus a short calibrated corrector should dominate long energy-sampling chains on physical quality, diversity, and cost.

4.4 MECHANISTIC BOLTZMANN ALIGNMENT

For held-out parent molecules and perturbation families, we measure the within-parent correlation and calibration slope between $\log p_\theta^{\text{pos}}(r \mid c)$ and $-\beta_{\text{phys}} E(r, c)$. We report this with OMol25 as a mechanism diagnostic, and repeat it with xTB and a DFT subset to test transfer.

4.5 TEMPERATURE ABLATION

We sweep the joint drift / bridge-final temperature $k_B T \in \{0.02569, 0.05, 0.10, 0.25\}$ eV to quantify the cost of training at a *tempered* Boltzmann bridge instead of at the physical anchor. Each row holds every other hyperparameter fixed at the v1 base config and only overrides `temperature.bridge_kT_final_eV` and `temperature.drift_kT_eV` to the row’s value (the 0.25 eV row additionally sets `bridge_kT_mode = fixed_tempered`, making the anneal a no-op). Physical-quality numbers use BGFM-Native-det at sampling time; Boltzmann-alignment numbers use BGFM-Native-300K.

Table 3: Temperature ablation. Higher Boltzmann slope (within-parent Pearson against $-\beta_{\text{phys}} E_{\text{NP}}$) is better; lower xTB ΔE is better.

$k_B T$ [eV]	Boltz. slope	xTB med. ΔE	Valid	Vendi	VULS
0.02569 (physical)	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
0.05 (tempered, mild)	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
0.10 (tempered, moderate)	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
0.25 (tempered, warm; fixed)	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]

4.6 ABLATIONS AND NEGATIVE CONTROLS

A valid physical improvement should disappear or weaken under corrupted labels. We therefore train or evaluate controls with shuffled forces, shuffled energies, wrong parent grouping, wrong temperature, force norm only, and random scalar regularization.

Table 4: Negative controls separating correct physical supervision from arbitrary regularization.

Model/control	OMol25 align.	xTB med. ΔE	DFT subset	Valid	Vendi
FM baseline	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
BGFM true labels	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
Shuffled force	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
Shuffled energy	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
Wrong parent grouping	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]
Wrong kT ($10 \times k_B T_0$)	[RUN:]	[RUN:]	[RUN:]	[RUN:]	[RUN:]

4.7 CROSS-MODEL RANKING AND FILTERING

We collect samples from EDM/FlowMol3/EBMol/BGFM and deliberately strained perturbations. We compare EBMol energy, OMol25 energy, BGFM’s calibrated energy head, and the BGFM combined score as predictors of independent xTB/DFT relaxation energy and high-strain failure. Metrics are Pearson, Spearman, AUROC for high-strain detection, and top-decile filtering quality.

5 LIMITATIONS

BGFM inherits OMol25’s coverage and accuracy limitations. The main density loss is conditional-coordinate unless the optional discrete joint-density extension is explicitly enabled. The short corrector is a learned-energy refinement, not an exact MCMC sampler. The energy head is a calibrated scorer distilled from neural-potential energies and forces, not a free-energy estimator. These limitations are evaluated with independent oracles and shuffled-label controls rather than hidden.

6 CONCLUSION

BGFM is designed to bridge two strengths: amortized de novo flow sampling and physically meaningful energy landscapes. By calibrating a fast molecular flow proposal with an external universal neural potential and using a short learned-energy corrector, BGFM targets low-strain, diverse, computationally efficient 3D molecule generation. The decisive empirical question is not whether BGFM improves an OMol25 self-metric, but whether it improves independent physical quality and the quality-diversity-compute frontier against strong flow and energy-based baselines.

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